

II. Strategy

1. Strategic analysis and actions

Twenty years after the stocklisting of Salzgitter AG (SZAG), securing the future of our Group remains at the heart of our entrepreneurial responsibility – for our shareholders, our employees, our places of business and society in which we are firmly embedded as a company. SZAG subscribes to a responsible corporate policy aimed at growing the value of the company. The management of our Group is based on values, and its strategy is designed for the long term. All decisions pursue the aim of preserving the substance of the Group and continually raising its performance. At the same time we do not lose sight of our social responsibilities. Strengthening our production facilities with due regard for the statutory and economic framework conditions forms the foundation of a good, target-oriented working relationship with our stakeholders. We are aware that from time to time conflicts of interest may arise between or even within individual stakeholders. The preservation of our competitiveness and the profitability of our entrepreneurial endeavors are essential prerequisites in order to act sustainably and responsibly in the sphere of Corporate Social Responsibility (CSR). In the long run, these objectives can only be achieved if sustainability criteria for mankind and the environment do not end at the factory gate or national borders, but all market players act on the same or at least similar competitive terms. As has been demonstrated once again, especially by the recent discussions, for instance, about steel imports to the EU or amendments to emissions trading from 2021, production with the highest environmental standards can only be sustainably guaranteed if the economic viability and preservation of jobs can be ensured in our own company. We recognize that European and German views of CSR topics are not given the same weight per se by any global yardstick in terms of their importance. As an internationally active Group, we must exercise cultural tolerance accordingly. We understand full well that global issues in particular, such as climate protection or respect for human rights, ultimately require a global framework in order to actively and productively shape the future. As a company, we can only play a limited role in influencing any such framework; it is in the nature of things that ultimately the onus lies with the politicians.

Our business activities benefit primarily from our own performance. At the same time, we must acknowledge that a reliable social framework, fair collaboration with partners based on trust and our relationship with our natural surroundings are of considerable importance. This obliges all employees of the Salzgitter Group to observe all laws anywhere and at any time when acting on behalf of the company, to respect generally accepted basic values when dealing with other people and other companies, to maintain fairness in our actions and to preserve nature to the best of our ability – or to put it in a nutshell: to act responsibly at all times. This is the starting point for the Salzgitter Group's Code of Conduct dated August 2015, which also declares the basic principles listed in the United Nations Global Compact to be the basis for actions taken in the company. Further information on the Code of Conduct is to be found both in Chapter III. "Process Management", 3. "Rules and Processes", and on our homepage. CSR is firmly established at SZAG. A CSR working group has been set up and located at Group level. It comprises the responsible managers of the relevant technical departments, and it reports directly to the Executive Board. This enables us to ensure that all aspects regarded as correct and important are incorporated into the corresponding specifications for the various departments via Group directives, instructions and target agreements.

However, we can only put CSR into practice successfully if all employees embrace the subject in the daily business of our Group companies.

For this reason, aspects such as environmental protection and occupational safety are components of target agreements with our managers alongside financial indicators. For example, CSR-related targets – in areas such as energy efficiency or health and safety at work – are defined for variable components of remuneration.

There is no separate sustainability strategy for the company as a whole. However, we draw up rules for aspects with relevance to the subject of sustainability via specific Group directives and other binding specifications directed at the workforce. Examples to be mentioned here are the Group directives on environmental protection and Corporate Compliance as well as the Code of Conduct issued by the Executive Board.

2. Materiality

Steel has been and still ranks as the number 1 construction material worldwide. It is a component of many industrial value chains – or even their starting point – and thereby contributes to the prosperity of those working in them.

At the same time, steel production and steel processing are necessarily associated with the consumption of relevant resources and energy, and particular attention must be paid to this aspect. A sharp eye must also be kept on working conditions in order to avoid any harmful effects for staff or areas surrounding production facilities.

Steel production and steel processing, together with associated activities, remain at the heart of the entrepreneurial value chain in the Salzgitter Group. We are therefore aware of our special responsibility for ensuring that our work is sparing on resources and energy efficient and for providing a high degree of workplace health and safety. With this in mind, priority is given to the Group's steel-related activities as defined by the criterion of "materiality" in the environmental sphere, as these activities far outweigh all other Group operations as far as the use of resources, consumption of energy and emissions are concerned. This process reveals any circumstances of material importance for obtaining a general understanding of the Salzgitter Group's business operations in this context. At the same time, it serves the purpose of defining the reporting framework and highlighting significant information. The findings from an internal materiality analysis and the results of stakeholder interviews held in September/October 2018 basically support the accuracy of these previous ideas (see also Chapter III. "Process Management", 5. "Stakeholder participation"). However, in addition to this observation, it should also be noted that, in accordance with materiality as defined by the CSR Directive Implementation Act (CSR-RUG), in a few places, the contents of the 2017 Non-Financial Report have now been classified as not relevant and have therefore not been elaborated on again in the 2018 Non-Financial Report. We also provide explanations of points portrayed in this Non-Financial Report for other reasons. We regard the aspects analyzed in more detail as relevant with respect to their possible effect on our own business operations. For example, analysis of the Group's total CO₂ emissions from operating its plants shows that around 90% is accounted for by the steel production of Salzgitter Flachstahl GmbH alone. This correlates above all with further air emissions as well as the consumption of resources and generation of waste. You can find further information on this in Chapter IV. "The Environment".

As far as employee matters are concerned, the following issues were also rated as material for the preparation of this year's Non-Financial Report: transparent and fair principles of remuneration on which we report in Chapter III. "Process Management", 4. "Incentive Systems"; employee rights both with regard to information and the workforce's participation and also with respect to the inclusion of employee interests in decision-making processes, as well as the set of issues surrounding the development of junior staff, personnel growth and training and further education which we cover in Chapter V. "Society", 3. "Qualifications".

In addition, we voluntarily report on our human resources strategy with the aid of which we respond to the challenges of demographic changes and that we portray on the basis of selected areas of our project "GO – The Generation Offensive 2025".

In addition, we report more comprehensively on the topics of human rights, political dialog as well as conduct compliant with laws and directives.

3. Targets

Salzgitter AG (SZAG) subscribes to qualitative and quantitative sustainability targets that can be measured over time. However, due to the complexity of our production systems and the products manufactured in them, it is usually not useful to attach absolute or relative targets across the Group to certain aspects. This is evidenced by the fact that we do not run any groupwide management systems, for instance in accordance with DIN EN ISO 14001 (environment) or DIN EN ISO 50001 (energy) standards, but implement them at Group company level depending on requirements. We will therefore focus below on individual targets in relevant areas of sustainability:

The environment/energy

- Energy savings in the Salzgitter Group
In line with the target of SZAG's energy efficiency network, a saving of 100,000 MWh from August 2016 to August 2019 through the implementation of corresponding efficiency measures throughout the Group.
- Heat treatment line investment project of Ilsenburger Grobblech GmbH
With the new construction started in 2018, 200,000 t of plate steel are to be refined in energy-efficient processes from the year 2020. The use of modern technology prevents significant loss of exhaust heat – by comparison with current technology – and allows a substantial reduction in the volume of material transports by rail to be achieved by moving the heat treatment line from Salzgitter to Ilsenburg.
- Software-based management system
Special software was introduced in the wake of conversion of the Integrated Management System of Salzgitter Flachstahl GmbH to meet the new requirements of the revised standards DIN EN ISO 14001, DIN EN ISO 9001 and IATF 16949. The aim is to make internal interfaces in particular – but also cross-company interfaces – more transparent from 2019, enabling existing synergies to be more effectively exploited.

Work/social issues

- Group mission statement “YOUNITED”
Establishment within the workforce of the six values behind the Group mission statement “YOUNITED” through annual initiatives promoting the “Value of the year”.
- Groupwide talent pool
Identification of talent across the Group with the objective of enhancing the Group's attractiveness as an employer by taking into account groupwide career prospects.
- Systematic succession planning
Systematic succession planning ensured by guaranteeing the provision of the necessary personnel resources over the medium and long term.
- Occupational safety
Reduction of absenteeism rates through measures and campaigns on occupational health and safety.

Energy consumption ¹⁾

		2018	2017
Primary energy sources²⁾	GWh	4,411	4,077
Energy used to generate steam and electricity from waste gas and waste heat	GWh	1,987	1,928
Electricity consumption	GWh	2,663	2,682
of which external power supply	GWh	1,227	1,315
Total energy consumption	GWh	5,638	5,392

¹⁾ In terms of materiality additional Group companies of Trading Business Unit, Technology Business Unit and EUROPIPE sites were considered for the energy consumption. In detail: EUROPIPE GmbH, Berg Pipe Mobile Corporation, Berg Pipe Panama City Corporation, GESIS Gesellschaft für Informationssysteme mbH, Glückauf Immobilien GmbH, Hansaport Hafenbetriebsgesellschaft mbH, Klöckner Desma Elastomertechnik GmbH, DESMA Schuhmaschinen GmbH, KHS GmbH, MÜLHEIM PIPECOATINGS GmbH, Salzgitter Automotive Engineering GmbH & Co. KG, Salzgitter Hydroforming GmbH & Co. KG, Salzgitter Mannesmann Forschung GmbH, TELCAT KOMMUNIKATIONSTECHNIK GmbH, TELCAT MULTICOM GmbH, Verkehrsbetriebe Peine-Salzgitter GmbH, VPS Infrastruktur GmbH, KHS Coroplast GmbH, Salzgitter Mannesmann Handel GmbH, Salzgitter Mannesmann Stahlhandel GmbH, Stahl-Center Baunatal GmbH, Universal Eisen und Stahl GmbH.

²⁾ Without reducing agents, they are considered under "Materials used".

Water consumption

		2018	2017
Total water consumption	Thousand cubic meters	18,614	18,493
of which drinking water	Thousand cubic meters	2,523	2,693
of which industrial water/service water	Thousand cubic meters	16,091	15,800

Waste water

		2018	2017
Total waste water	Thousand cubic meters	13,749	15,877
of which direct discharge	Thousand cubic meters	13,211	15,315
of which indirect discharge	Thousand cubic meters	538	562

While the volume of water consumed in the current reporting year was in a similar range to 2017, waste water volumes for 2018 were appreciably lower than for the previous year. This is basically due to very low rainfall and high evaporation rates in the comparatively dry, warm summer of 2018.

Waste

		2018	2017
Total waste	kt	1,231	1,011
of which recycled	kt	798	730
Hazardous waste as a share of the total	kt	62	55

There was a noticeable increase in the amount of waste generated in 2018. The majority of it results from construction work in connection with the current "Heat Treatment Line" investment project at Ilsenburger Grobblech GmbH (see also Chapter II. "Strategy", 3. "Objectives"). Disposal of the excavated soil / rubble generated is leading to higher numbers in the key waste indicators shown.

Air emissions¹⁾

		2018	2017
NO _x	t	4,839	4,730
SO _x	t	4,692	5,450
Particulate matter (PM10)	t	162	183

¹⁾ Facilities for which an emission declaration is made pursuant to Section 11 of the German Federal Control Act (BImSchV) or which are listed in the PRTR register.

Lower SO_x emissions were achieved in 2018 than in 2017 as a result of technical and operational measures such as optimizing process control in Salzgitter Flachstahl GmbH's sinter plant. A reduction in emissions was also recorded for the parameter of fine dust particles. This result is primarily based on operating and technical factors such as lower operating hours, that as well as unavoidable variations caused by the method used which always play a role in the measurement of all indicators.

3. Emissions impacting climate

Salzgitter AG (SZAG) participated in the CDP reporting format (formerly Carbon Disclosure Project) for the first time in 2016 and subsequently in 2017 and 2018. For its initial participation in 2016, SZAG was crowned "Best Newcomer Germany" by CDP. While we achieved a "B" rating in 2016 and 2017, we managed to further improve our CDP performance in 2018 and for the first time we were awarded an "A-" rating.

CDP is an international not-for-profit organization. It records and evaluates the greenhouse gas emissions (GHG emissions) voluntarily reported by companies and organizations as well as the way in which opportunities and risks resulting from climate change are handled and strategies evolved in response. Once a year, CDP collects extensive data on behalf of investors, and it claims to have now established the most important data-base and analysis on the subject in the world.

Based on the system of the Greenhouse Gas Protocol which is also used by CDP, we report GHG emissions in the categories Scope 1 to Scope 3 as shown below. The definitions are as follows.

- Scope 1 = direct GHG emissions from operating plants
- Scope 2 = indirect, energy-related GHG emissions
- Scope 3 = further indirect GHG emissions from the purchase of raw materials

We have used the specifications of the Greenhouse Gas Protocol Corporate Standard for calculating the greenhouse gas emissions in Scope 1 to Scope 3. The majority of our direct emissions are also subject to the European Emissions Trading System (ETS) with the result that the strict specifications contained in the Monitoring Regulation are applied in this area. The data for Scope 1 and 2 result from energy consumption figures shown in Section 2 "Resources Management" as well as details from ETS monitoring.

As an integrated steel mill, Salzgitter Flachstahl GmbH (SZFG) occupies a special position with regard to GHG emissions, and it therefore operates its own energy savings scheme and GHG emissions reduction program. In 2017, these programs achieved a cumulative reduction in Scope 1 and Scope 2 emissions of 190 kt CO₂ as a direct consequence of efficiency measures. In 2018, the figure was 208 kt CO₂.

GHG emissions

		2018	2017
Direct GHG emissions from operating plants (Scope 1)	kt CO ₂	8,710	8,470
Indirect GHG emissions (Scope 2)	kt CO ₂	600	693
Other indirect GHG emissions from feedstock purchase ¹⁾ (Scope 3)	kt CO ₂	1,508	1,362
Reduction of GHG emissions as a direct result of efficiency measures ²⁾	kt CO ₂	208	190

¹⁾ Without reducing agents; specific emission factors based on supplier information or common available sources. Survey is oriented to Corporate Value Chain Standard (Scope 3) of the Greenhouse Gas Protocol Initiative of the World Resources Institute (WRI) and the World Business Council for Sustainable Development (WBCSD)

²⁾ Resulting from measures of the EE project of Salzgitter Flachstahl GmbH, affecting Scope 1 and Scope 2.

The reduction in Scope 2 emissions in the current reporting year by comparison with the previous year is due both to lower volumes of external power procured (see also IV “Environment” 2 “Resource Management”) and to the lower average carbon dioxide emission factor in the German electricity mix. The higher Scope 3 emissions in 2018 result from increased use of lime at SZFG as well as higher emissions from pellet production at the relevant suppliers.

The greatest challenge for the Group in connection with emissions affecting the climate is the fact that CO₂ emissions based on technologies development largely today on an international scale and in the Salzgitter Group have nearly reached their minimum possible level within the process boundaries set by laws of nature. There are therefore strict limits to future reductions in greenhouse gas emissions achievable merely through efficiency measures. SZAG therefore set up a project named SALCOS® (Salzgitter Low Carbon Steelmaking) in 2015 together with the Fraunhofer Gesellschaft. The aim is to examine and pursue a gradual transformation from today’s carbon-based to hydrogen- and electricity-based steel production. The measures derived from the project are intended to be implemented in the Salzgitter steel works for the successive, stepwise conversion of production.

In order to achieve a significant degree of decarbonization in its primary steelmaking, SZAG wants to exploit a unique feature of iron metallurgy: Hydrogen can replace carbon in the reduction of iron ore, which leads to the formation of water (H₂O) instead of carbon dioxide (CO₂). The process heat required in steelmaking and further processing can also be provided by electrical energy instead of carbon-based sources. This means that ultimately nearly all CO₂ emissions caused by steel production can be avoided.

To make a clear distinction between our new ideas and the concepts of CO₂ usage (Carbon Capture and Usage, CCU) or storage (Carbon Capture and Storage, CCS), we coined the expression “Carbon Direct Avoidance” (CDA) in November 2016. The purpose of this neutral acronym is to show that our approach to the problem is not restricted to SZAG, but is technically applicable in principle at all integrated steelmaking facilities around the world.

In the course of the project, the preconditions and impact of the proposed implementation steps on our Group’s integrated steel mill in Salzgitter will be examined. Scientific support comes from the neutral expertise of “Fraunhofer Gesellschaft”, one of the leading research organizations for applied science in Europe.

If SALCOS® is implemented, all technical measures required for this technology are to be realized at the Salzgitter facility in order to safeguard our company’s existing production plants and jobs as far as possible and prevent any carbon leakage to areas outside the ETS.

The project is based on a modular concept offering the possibility of gradual implementation. The steps involved can adapt to future challenging CO₂ reduction targets in Europe after 2030. Energy-efficient steelmaking at the integrated steel mill in Salzgitter currently in existence has to be extensively modified on the basis of the SALCOS® concept, as outlined below.

The combination of direct reduction technology and electrolysis technology permits the optimum use of volatile renewable energies as, depending on their availability, electrolytic hydrogen can flexibly replace natural gas in the direct reduction process, thus also allowing for storing of renewable energy in excess production of directly reduced iron, so-called DRI.

It must be emphasized that combining such a flexibly operated direct reduction plant with an existing integrated steel mill has never been implemented anywhere in the world, and thus represents a major challenge primarily in terms of process integration but also from an economic perspective. As is the case with alternative concepts to reduce CO₂ emissions in the steel industry, the deployment of SALCOS® within the given regulatory and economic framework conditions could lead to an increase operating costs and jeopardize competitiveness. Against this background, corresponding framework conditions must be created which enable the proposed project to be realized.

The major strength of SALCOS® is that its approach is based on industrially established technology (gas-based direct reduction) that can be adapted in response to future challenges. From a technical perspective, this offers the chance to achieve a significant reduction in CO₂ emissions early, thus not requiring any tedious and costly laboratory facilities, by using a plant on an industrial scale. It is our conviction that we must act now if economy and society wish to achieve the climate targets set for 2050. This is where SALCOS® clearly differs from many other decarbonization approaches, most of which mainly can only be realized on an industrial scale from 2035 onwards.

If production is completely converted and only hydrogen is used for iron reduction, the maximum reduction in CO₂ emissions achievable with the SALCOS® concept is around 95%.

As with comparable approaches, SALCOS® will require very high investment in new plant equipment which will be in the order of several billion euros if it is fully implemented. In this context, the following points should be noted: For the foreseeable future, techniques for reducing CO₂ emissions will not offer any higher energy efficiency, lower energy costs or better product quality than the status quo that might lead to lower production costs or higher sales. The added value lies solely in steelmaking with reduced CO₂ emissions in order to fulfil the task that society has set us. The domestic steel industry must respond locally to the challenge of reducing CO₂ emissions while continuing to hold its own against global competition. We are therefore applying for considerable sums of public funding in order to implement SALCOS®.